DAY - 12 SEAT NUMBER | | | | | | |

2013	III	08	1100	V – 734	(E)

COMPUTER SCIENCE
PAPER - I (D-9)

Time : 3 Hours	4 Pages	Max. Marks : 50

Instructions : *(1) All questions are compulsory.*

(2) Figures to the right indicate full marks.

(3) Use of any type of calculator is not allowed.

(4) Header files and comments in C++ program are not necessary.

(5) Draw neat diagram wherever necessary.

1. (A) Select correct alternatives and rewrite the following : 2

(a) The time required to execute the bubble sort algorithm is proportional to _____. 1

(i) n

(ii) 2n–1

(iii) n^2

(iv) $\frac{n+1}{2}$

(b) The long form of HTML is _____. 1

(i) Hyperlink Tag Markup Language

(ii) Hyper Text Markup Language

(iii) Hypermedia Tag Markup Language

(iv) Hyperlink Text Markup Language

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(d) If the page size of 1MB Memory is 1KB, then the number of higher order bits of the address bus used to denote page number is _____ 1

(i) 10

(ii) 11

(iii) 12

(iv) 9

(d) Object Oriented Programming follows _____ approach in program design. 1

(i) top-down

(ii) non-hierarchical

(iii) random

(iv) bottom-up

(B) Answer any one of the following : 3

(a) Explain in short the function of :

(i) Program Manager

(ii) File Manager

(iii) Control Panel

in the MS-Windows Environment.

(b) Explain Stack and Queue Data Structure with suitable example. 3

(c) Explain use of the following tags with example : 3

(i) <a>

(ii) <UL>

(iii) <TH>

2. (A) Answer any two of the following : 6

(a) State any five principal advantages of Object Oriented Programming. 3

(b) What is a Linked List ? Show a linked list with suitable example having five nodes with a properly labelled diagram. 3

(c) Explain Binary Search Algorithm with a suitable example. 3

(B) Answer any one of the following : 4

(a) What is a Video RAM ? Explain use of Attribute Byte and Data Byte of Video RAM with example. 4

(b) State any eight rules for operator overloading in C++. 4

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3. (A) Answer **any two** of the following :

(a) What is a Computer Worm ? Explain its Modes of Operation. 3

(b) What is a class in C++ ? Write general form of a Class Declaration. 3

(c) What is a Constructor in C++ ? State any four special characteristics of Constructor Function. 3

(B) Answer **any one** of the following :

(a) Explain in short the three main functions to be carried out by Memory Management Module of Operating System. State any two types of Contiguous Real Memory Management Systems. 4

(b) Draw Binary Tree Structure for the following expression :

$(2A + B) (5F - D)^3$ 4

4. (A) Answer **any two** of the following :

(a) With a suitable example, show representation of records in a Memory. 3

(b) Explain in brief, three major services provided by the Operating System. 3

(c) Explain in short the steps involved in the Process of Operator Overloading in C++. 3

(B) Answer **any one** of the following :

(a) What is a Process ? Explain running, ready and blocked process states of Process Management. 4

(b) What does Polymorphism means in C++ ? How is Polymorphism achieved at Compile Time and Run Time. 4

5. Answer **any two** of the following :

(a) Write a C++ program to find surface area and volume of a sphere for a given radius (r), using function computesphere ( ) that returns volume (v) and the surface area (a) of the sphere with given radius (r), void computesphere (float & v, float & a, float r). 5

(b) Write a C++ program to find largest of four given integers using max( ) function to return largest of four given integers :
int max (int, int, int, int). 5

(c) Write exact output of the following HTML Code with font specifications in brackets : 5

```
<html>
<body>
<h1 align = "center" > H.S.C. BOARD EXAMS </h1>
<p align = "center" > <i> SUBJECT : COMPUTER SCIENCE </i> </p>
<p align = "center" > <b> THEORY AND PRACTICAL EXAMS </b> </p>
<p align = "center" > <u> TOTAL </u> _____ 200 MARKS </p>
</body>
</html>
```

OR

5. Answer **any two** of the following :

(a) Write a program in C++ that inputs and stores 10 numbers in an array and prints numbers in reverse order. 5

(b) Write a C++ program to find smallest of four given integers using min( ) function to return smallest of the four given integers :

int min (int, int, int, int) 5

(c) Write HTML Code for the following : 5

HTML COLOUR CODE IN HEXA-DECIMAL	
BLACK	WHITE
# 000000	# FFFFFF

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